
ZETAPLEX SQT CONC

SEQUESTERING AGENT / HEAVY METAL CHELANT

Product Description:

Zetaplex SQT CONC is a highly efficient, high-temperature stable sequestering agent specially engineered for textile wet processing. It delivers powerful chelation of heavy metal ions including calcium (Ca^{2+}), magnesium (Mg^{2+}), iron (Fe^{3+}), copper (Cu^{2+}), and other metallic impurities that originate from process water, chemicals, or natural fibres. These heavy metal ions, if left uncontrolled, lead to metal salt precipitation, lower whiteness in bleached fabrics, uneven dyeing, loss of dye yield, and poor reproducibility.

Zetaplex SQT CONC effectively eliminates these risks, ensuring consistent, high-quality output across both batch and continuous processing operations. Its anionic, water-soluble liquid formulation enables easy handling and precise dosing, while its broad compatibility with acids, alkalis, and other process chemicals makes it a reliable choice across all pre-treatment stages including demineralisation, scouring, and bleaching.

Key Benefits:

Superior Iron Chelation: Provides high iron chelation properties, effectively controlling loose spots and pinhole problems that compromise fabric quality.

Excellent Demineralising Action: Performs efficiently in both exhaust (batch) and continuous processes, removing hardness-causing ions and metallic impurities from process water and fibres.

Water Quality Independence: Performance remains consistent regardless of variations in feed water quality or hardness levels, ensuring reliable and reproducible results.

Enhanced Whiteness in Bleaching: When used in the bleach bath, it delivers brighter and cleaner whites by preventing interference from heavy metal ions.

Scale Prevention: Eliminates scale build-up on bleaching machinery and process equipment, reducing maintenance downtime and extending machine life.

Suitable for Dyeing: When used in the dye bath, helps achieve brighter, more uniform shades by preventing metal-induced dye precipitation and unlevel dyeing.

Threshold Effect: Maintains clarity in supersaturated solutions of CaCO_3 , CaSO_4 , BaSO_4 , $\text{Fe}(\text{OH})_3$ at very low dosage levels — keeping sequestered metal ions in solution even in the presence of precipitating agents.

Reduced Ash Content: Effectively lowers ash content on fibre substrates, contributing to cleaner and higher-quality finished goods.

Properties:

Chemical Constitution	Based on organically modified compounds and organic acid chemistry
Ionic Character	Anionic
Appearance	Colourless to light straw yellow clear liquid
Physical Form	Liquid
Solubility	Freely soluble in water
Storage Stability	Stable under normal climate conditions; frozen material may be allowed to return to ambient temperature naturally
Compatibility	Compatible with most anionic and non-ionic surfactants, acids, and alkalis
Stability to Alkalis	Very Good
Stability to Acids	Very Good

Applications & Dosage:

Heavy metal ions — whether originating from process water, chemicals such as salt, soda ash, or alkali used in dyeing, or from natural fibres — can cause serious problems including:

- Lower whiteness of bleached fabric
- White spots and uneven tone on dyed substrates
- Reduction in dye yield and loss of reproducibility
- Unlevel dyeing, especially in yarn dyeing where tightly wound packages act as a filter
- Reduction in fastness properties
- Loss of "Right First Time" production

1. Demineralisation Bath — Exhaust Process (Batch)

Function	Product	Dosage	Parameters & Conditions
Sequestering Agent	ZETAPLEX SQT CONC	0.3–1.0 g/l	Auxiliary dosing – raise temperature to 60°C–70°C – hold for 10–15 minutes
Wetting Agent	Zetawet SWLF CONC	0.3–0.5 g/l	

2. Demineralisation Bath — Padding / Continuous Process

Function	Product	Dosage	Parameters & Conditions
Sequestering Agent	ZETAPLEX SQT CONC	1.0–2.0 g/l	Pad at 70°C – dwell in J-box for 20 minutes – followed by hot wash
Wetting Agent	Zetawet SWLF CONC	0.5–0.7 g/l	

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